

Equivalent definitions of Nichols algebras

Lukas Simons

February 2026



Contents

- Coalgebra theory
- Braided vector spaces
- Equivalent characterizations

Definition

A *coalgebra* (over k) is a triple (C, Δ, ε) with C a vector space, $\Delta: C \rightarrow C \otimes C$, and $\varepsilon: C \rightarrow k$ satisfying that

$$\begin{array}{ccccc}
 C & \xrightarrow{\Delta} & C \otimes C & & C \otimes k \xleftarrow{\text{id}_C \otimes \varepsilon} C \otimes C \xrightarrow{\varepsilon \otimes \text{id}_C} k \otimes C \\
 \downarrow \Delta & & \downarrow \text{id}_C \otimes \Delta & & \searrow \sim \quad \uparrow \Delta \quad \swarrow \sim \\
 C \otimes C & \xrightarrow{\Delta \otimes \text{id}_C} & C \otimes C \otimes C & & C.
 \end{array}$$

Example

If (A, μ, η) is a **finite-dimensional** algebra, then A^* is a coalgebra as

$$\Delta: A^* \xrightarrow{\mu^*} (A \otimes A)^* \xrightarrow{\sim} A^* \otimes A^*,$$

$$\varepsilon: A^* \xrightarrow{\eta^*} k^* \xrightarrow{\sim} k.$$

Definition

A vector space V is *graded* if it admits subspaces $V(n)$ such that

$$V = \bigoplus_{n \geq 0} V(n).$$

A *morphism of graded vector spaces* $V \rightarrow W$ is a linear map $f: V \rightarrow W$ such that $f(V(n)) \subseteq W(n)$ for all $n \geq 0$.

Definition

A coalgebra (C, Δ, ε) is *graded* if C is, and Δ and ε are graded maps. This is,

$$\Delta(C(n)) \subseteq \bigoplus_{i=0}^n C(i) \otimes C(n-i),$$

$$\varepsilon(C(n)) = \begin{cases} k, & \text{if } n = 0, \\ 0, & \text{else.} \end{cases}$$

Definition

A graded coalgebra C is *connected* if $C(0) \cong k$.

Proposition

A connected coalgebra admits a unique group-like element 1_C , this is, a non-zero $x \in C$ such that $\Delta(x) = x \otimes x$. Moreover, $1_C \in C(0)$.

Proof.

Let $x \in C(0)$ be non-zero, then as $\Delta(x) \in C(0) \otimes C(0)$, there is a unique non-zero $\lambda \in k$ such that $\Delta(x) = \lambda x \otimes x$. Hence λx is the unique group-like element of C . Assume the existence of a group-like element $x = \sum_{j=0}^N x_j$ for $x_j \in C(j)$ with $x_N \neq 0$. If $N > 0$, notice that the $2N$ -homogeneous component of $\Delta(x) = x \otimes x$ is $x_N \otimes x_N$. However, $\Delta(x) \in \sum_{j=0}^N \bigoplus_{r=0}^j C(r) \otimes C(j-r)$ which is disjoint from $(C \otimes C)(2N)$ as $N > 0$. Thus $N = 0$, which reduces it to the previous case. □

Definition

Let C be a connected coalgebra, the *primitive elements of C* are

$$\mathcal{P}(C) = \{x \in C \mid \Delta(x) = x \otimes 1_C + 1_C \otimes x\}.$$

Proposition

Let C be a connected coalgebra, then $C(1) \subseteq \mathcal{P}(C)$.

Proof. If $x \in C(1)$ then

$\Delta(x) = 1_C \otimes y + z \otimes 1_C \in C(0) \otimes C(1) + C(1) \otimes C(0)$. Thus by applying $\text{id} \otimes \varepsilon$ respectively $\varepsilon \otimes \text{id}$ it follows that $z = y = x$. $\square$

Definition

A *strictly graded coalgebra C* is a connected coalgebra such that $\mathcal{P}(C) \subseteq C(1)$, and thus $\mathcal{P}(C) = C(1)$.

If (C, Δ, ε) is a **graded coalgebra**, we define for $i_1 + \cdots + i_r = n$

$$\Delta_{i_1, \dots, i_r} : C(n) \xrightarrow{\Delta^{r-1}} C^{\otimes r} \xrightarrow{\pi_{i_1} \otimes \cdots \otimes \pi_{i_r}} C(i_1) \otimes \cdots \otimes C(i_r),$$

$$\Delta_{1^n} = \Delta_{1, 1, \dots, 1},$$

$$I_C(n) = \ker(\Delta_{1^n}) \subseteq C(n).$$

Theorem

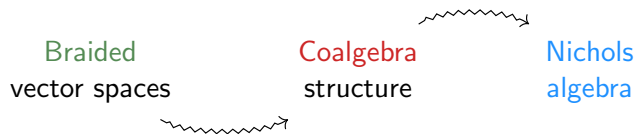
Let C be a connected coalgebra, then $I_C = \bigoplus_{n \geq 2} I_C(n)$ is a graded coideal of C . Moreover,

- ▶ *I_C is the largest coideal of C contained in $\bigoplus_{n \geq 2} C(n)$*
- ▶ *I_C is the unique coideal I of C contained in $\bigoplus_{n \geq 2} C(n)$ such that C/I is strictly graded.*

In particular, $\mathcal{B}(C) = C/I_C$ is a strictly graded coalgebra.

Proof. Heckenberger, Schneider [4, Theorem 1.6.13].





Definition

A *braided vector space* (V, c) is a vector space V , equipped with a linear isomorphism $c \in \text{End}(V \otimes V)$ such that

$$(c \otimes \text{id}_V)(\text{id}_V \otimes c)(c \otimes \text{id}_V) = (\text{id}_V \otimes c)(c \otimes \text{id}_V)(\text{id}_V \otimes c).$$

A *braided linear map* $f: (V, c) \rightarrow (W, d)$ is a linear map $f: V \rightarrow W$ such that

$$(f \otimes f)c = d(f \otimes f).$$

We denote the category of braided vector spaces as **BrVec**.

Definition

A *graded (co-/bi-/Hopf) algebra* is a graded vector space B with additional algebraic structure, such that the structure morphisms are graded.

Explicitly, for a bialgebra B

$$\begin{aligned} B(n)B(m) &\subseteq B(n+m), & 1_B &\in B(0), \\ \Delta(B(n)) &\subseteq \bigoplus_{i=0}^n B(i) \otimes B(n-i), & \varepsilon(B(n)) &= \begin{cases} k, & \text{if } n = 0, \\ \{0\}, & \text{else.} \end{cases} \end{aligned}$$

Given a morphism $f: V \rightarrow W$ and $0 \leq m \leq n-1$, we denote

$$f^{\uparrow m} = \text{id}_{V^{\otimes m}} \otimes f \otimes \text{id}_{V^{\otimes n-m-1}}.$$

Definition

Let (V, c) be a braided vector space, and $n, m \geq 1$, then we define the morphism $V^{\otimes n} \otimes V^{\otimes m} \rightarrow V^{\otimes m} \otimes V^{\otimes n}$

$$\begin{aligned} c_{n,m} &= (c_m c_{m-1} \cdots c_1)(c_m c_{m-1} \cdots c_1)^{\uparrow 1} \cdots (c_m c_{m-1} \cdots c_1)^{\uparrow n-1} \\ &= (c_1 c_2 \cdots c_n)^{\uparrow m-1} (c_1 c_2 \cdots c_n)^{\uparrow m-2} \cdots (c_1 c_2 \cdots c_n). \end{aligned}$$

Definition

Let (V, c) be a braided vector space, $n, m \geq 0$, and $f: V^{\otimes n} \rightarrow V^{\otimes m}$, then f commutes with the braiding of V if

$$(f \otimes \text{id}_V)c_{1,n} = c_{1,m}(\text{id}_V \otimes f), \quad (\text{id}_V \otimes f)c_{n,1} = c_{m,1}(f \otimes \text{id}_V).$$

Definition

Given a braided vector space (V, c) , the category $\mathcal{C}(V, c)$ consists of tensor powers of V up to isomorphism (this is $V^{\otimes n} = V^{\otimes m} \otimes V^{\otimes n-m}$ for all $0 \leq m \leq n$), and morphisms being linear maps commuting with the braiding of V . The composition is this of the underlying linear maps.

$\mathcal{C}(V, c)$ is monoidal for the monoidal structure inherited from Vec , and braided by $\bar{c}_{V^{\otimes n}, V^{\otimes m}} = c_{n,m}$.

In order to differentiate between the monoidal product of vector spaces and in $\mathcal{C}(V, c)$, we will denote the latter's by $\underline{\otimes}$.

Definition

A *braided graded (co-/bi-/Hopf) algebra* is a braided and graded vector space (B, d) with additional algebraic structure, such that the structure morphisms are graded and commute with its braiding, and moreover

$$d(B(i) \otimes B(j)) = B(j) \otimes B(i) \text{ for all } i, j \geq 0.$$

Alternatively, **braided algebraic structures** on (V, c) are algebraic structures in the **braided monoidal category** $(\mathcal{C}(V, c), \underline{\otimes}, \bar{c})$.

This allows us to define braided graded algebraic structures as graded algebraic structures in said category with an additional compatibility.

Theorem

Let (V, c) be a braided vector space, then we may construct the free braided graded Hopf algebra $T(V, c) = (T(V), \bar{c})$, this is the graded vector space $T(V) = \bigoplus_{n \geq 0} V^{\otimes n}$ with the concatenation, and comultiplication defined on $V = T^1(V)$

$$\Delta(x) = x \otimes 1 + 1 \otimes x.$$

This satisfies the universal property that for any braided graded bialgebra (B, d) and $f: (V, c) \rightarrow (B, d)$ braided such that $\text{im}(f) \subseteq \mathcal{P}(B) \cap B(1)$, there is a unique braided graded bialgebra morphism $\bar{f}$ extending f ;

$$\begin{array}{ccc} (V, c) & \xrightarrow{f} & (B, d) \\ \downarrow & \nearrow \bar{f} & \\ (T(V), \bar{c}) & & \end{array}$$

Definition

The Nichols algebra of (V, c) is the Nichols algebra of the connected coalgebra $T(V, c)$;

$$\mathcal{B}(V, c) = \frac{T(V, c)}{I_V} = k \oplus V \bigoplus_{n \geq 2} \frac{V^{\otimes n}}{\ker(\Delta_{1^n})}.$$

$I_V = \bigoplus_{n \geq 2} \ker(\Delta_{1^n})$ is a **braided and graded** subspace of $T(V, c)$, and hence $T(V, c)/I_V$ a graded braided vector space.

Theorem

Let (V, c) be a braided vector space, then I_V is a braided graded Hopf ideal of $T(V, c)$. In particular, $\mathcal{B}(V, c)$ is a strictly graded braided Hopf algebra.

Proof. One verifies that $T(V)I_V$ and $I_V T(V)$ are coideals of $T(V)$, whence I_V is a braided graded biideal. Likewise, $S(I_V)$ is a coideal of $T(V)$, proving the claim. $\square$

Braided symmetrizers

Definition

Let (V, c) be a braided vector space, the *braided symmetrizers* are defined as $(n \geq 2)$

$$\Omega^2 = \text{id} + c,$$

$$\Omega^{n+1} = (\text{id} \otimes \Omega^n)(\text{id} + c_1 + c_1 c_2 + \cdots + c_1 c_2 \cdots c_n).$$

Theorem

Let (V, c) be a braided vector space, then for $n \geq 2$

$$\Omega^n = \Delta_{1^n}.$$

In particular,

$$\mathcal{B}(V, c) = T(V, c) / \ker(\Omega) = k \oplus V \bigoplus_{n \geq 2} V^{\otimes n} / \ker(\Omega^n).$$

Sketch of proof. We utilize that

- ▶ $\Delta_{1^n} = (\Delta_{1^i} \otimes \Delta_{1^{n-i}}) \Delta_{i, n-i},$
- ▶ for $x_1 \otimes \cdots \otimes x_n \in V^{\otimes n},$

$$\Delta(x_1 \otimes \cdots \otimes x_n) = \prod_{i=1}^n (x_i \underline{\otimes} 1 + 1 \underline{\otimes} x_i).$$

The result follows by induction.



Approximations

Definition

Let (V, c) be a braided vector space and (B, d) a braided graded Hopf algebra. Then (B, d) is a *pre-Nichols algebra* of (V, c) if

1. $B(0) = k$,
2. $B(1) \cong V$ as braided vector space,
3. $B(1)$ generates the algebra B .

Proposition

(B, d) is a pre-Nichols algebra if and only if there exists a braided graded Hopf ideal I of $T(V, c)$ contained in $\bigoplus_{n \geq 2} T^n(V)$ such that $(B, d) \cong T(V, c)/I$.

Definition

Let (V, c) be a braided vector space and (B, d) a braided graded Hopf algebra. Then (B, d) is a *post-Nichols algebra* of (V, c) if

1. $B(0) = k$,
2. $B(1) \cong V$ as braided vector space,
3. B is strictly graded, this is $\mathcal{P}(B) = B(1)$.

Theorem

$(B, d) \cong \mathcal{B}(V, c)$ if and only if (B, d) is a pre-Nichols and post-Nichols algebra of (V, c) .

Proof. $\mathcal{B}(V, c)$ is pre- and post-Nichols. If (B, d) is pre-Nichols then $(B, d) \cong T(V, c)/I$ for some braided graded Hopf ideal I contained in $\bigoplus_{n \geq 2} T^n(V)$. Since this is moreover strictly graded as it is post-Nichols, $I = I_V$ and $(B, d) \cong \mathcal{B}(V, c)$. $\square$

Bilinear forms

Definition

Let V, W be k -vector spaces, and $\langle \cdot, \cdot \rangle: V \times W \rightarrow k$ a bilinear form. The *extended form of $\langle \cdot, \cdot \rangle$* is the unique linear map

$$(\cdot, \cdot): T(V) \otimes T(W) \rightarrow k$$

satisfying that

1. $(1, 1) = 1$,
2. $(T^n(V), T^m(W)) = 0$ for all $n \neq m$,
3. For all $v_1 \otimes \cdots \otimes v_n \in T^n(V)$, $w_1 \otimes \cdots \otimes w_n \in T^n(W)$

$$(v_1 \otimes \cdots \otimes v_n, w_1 \otimes \cdots \otimes w_n) = \prod_{i=1}^n \langle v_{n-i+1}, w_i \rangle.$$

Lemma (A)

- i) If $\langle , \rangle$ is non-degenerate, then $(,)$ is non-degenerate.
- ii) If $\langle , \rangle$ is non-degenerate, V and W are finite-dimensional, and (V, c) is a braided vector space, so too is (W, d) for $d: W \otimes W \rightarrow W \otimes W$ defined by

$$(v, d(w)) = (c(v), w)$$

for $v \in V \otimes V, w \in W \otimes W$.

Theorem (B)

1. *If $(c(v), w) = (v, d(w))$, there is a unique well-defined linear map $[,]: T(V) \otimes T(W) \rightarrow k$ such that*

- ▶ $[1, 1] = 1$,
- ▶ $[T^n(V), T^m(W)] = 0$ for $n \neq m$,
- ▶ For $v \in T^n(V)$, $w \in T^n(W)$,

$$[v, w] = (\Omega^n(v), w) = (v, \Omega^n(w)).$$

2. *If $\langle , \rangle$ is non-degenerate and $(c(v), w) = (v, d(w))$, then*

$$\mathcal{B}(V, c) = T(V) / \ker([\bullet, T(W)]),$$

$$\mathcal{B}(W, d) = T(W) / \ker([T(V), \bullet]).$$

Proof. Heckenberger, Schneider [4, Theorem 7.2.3].



Proposition

Let (V, c) be a finite-dimensional braided vector space, then

- 1. the evaluation $\langle | \rangle: V \otimes V^* \rightarrow k$ is non-degenerate,*
- 2. (V^*, c^*) is a braided vector space,*

Thus

$$\mathcal{B}(V^*, c^*) = T(V^*) / \ker([T(V), \bullet]),$$

$$\mathcal{B}(V, c) = T(V) / \ker([\bullet, T(V^*)]).$$

Proof. The evaluation $\langle | \rangle$ is indeed non-degenerate, and hence (V^*, c^*) is a braided vector space by Lemma (A). By construction the conditions of Theorem (B) are satisfied. □

Skew derivations

Definition

Let $C = \bigoplus_{n \geq 0} C(n)$ be a graded coalgebra. For any linear functional $f \in C(1)^*$ we define the linear maps

$$\partial_f^l: C \rightarrow C: x \mapsto (f \otimes \text{id})(\pi_1 \otimes \text{id})\Delta(x),$$

$$\partial_f^r: C \rightarrow C: x \mapsto (\text{id} \otimes f)(\text{id} \otimes \pi_1)\Delta(x).$$

Definition

Let C be a graded coalgebra. A linear subspace $I \subseteq C$ is said to be ∂^l -invariant (respectively ∂^r -invariant) if $\partial_f^l(I) \subseteq I$ (respectively $\partial_f^r(I) \subseteq I$) for all $f \in C(1)^*$.

Given a **graded coalgebra** C , denote $C^+ = \bigoplus_{n \geq 1} C(n)$.

Proposition

Let C be a connected graded coalgebra.

- 1. If C is strictly graded and $x \in C^+$, then $x = 0$ if and only if $\partial_f^l(x) = 0$ for all $f \in C(1)^*$.*
- 2. I_C is the largest ∂^l -invariant subspace of C^+ .*

Proof. Heckenberger, Schneider [4, Proposition 7.3.3].



Corollary

Let C be a connected coalgebra, and $x \in C^+$. Then $\pi(x) \in \mathcal{B}(C)$ is non-zero if and only if there exists a linear functional $f \in C(1)^$ such that $\partial_f^l(x) \notin I_C$.*

Nichols algebra functor

Definition

The category of *strictly graded braided Hopf algebras* consists of

- ▶ objects $(H, c, \mu, \eta, \Delta, \varepsilon, S)$ being strictly graded braided Hopf algebras,
- ▶ morphisms being braided and graded Hopf algebra morphisms.

We denote this category by $\mathfrak{C}$.

Theorem

Let (V, c) and (W, d) be braided vector spaces and $f: (V, c) \rightarrow (W, d)$ a braided linear map, then there is a unique morphism of strictly graded braided Hopf algebras

$$\mathcal{B}(f): \mathcal{B}(V, c) \rightarrow \mathcal{B}(W, d)$$

such that the following diagram commutes;

$$\begin{array}{ccccc}
 (V, c) & \xrightarrow{\quad f \quad} & (W, d) & & \\
 \downarrow p_V \circ i_V & \searrow i_V & \swarrow i_W & \downarrow p_W \circ i_W & \\
 & T(V, c) & & T(W, d) & \\
 & \swarrow p_V & & \searrow p_W & \\
 \mathcal{B}(V, c) & \xrightarrow{\quad \mathcal{B}(f) \quad} & \mathcal{B}(W, d) & &
 \end{array}$$

Proposition

Let (V, c) , (W, d) , and (R, e) be braided vector spaces, then

- 1. $\mathcal{B}(\text{id}_{(V, c)}) = \text{id}_{\mathcal{B}(V, c)}$,*
- 2. for all braided linear maps $f: (V, c) \rightarrow (W, d)$ and $g: (W, d) \rightarrow (R, e)$, $\mathcal{B}(g \circ f) = \mathcal{B}(g) \circ \mathcal{B}(f)$.*

Corollary

The Nichols algebra functor

$$\mathcal{B}: \text{BrVec} \rightarrow \mathfrak{C}$$

is a well-defined functor.

For any **strictly graded braided Hopf algebra** (H, d) , we denote

$$\mathfrak{R}(H, d) = (H(1), \tilde{d}), \quad \text{for } \tilde{d} = d|_{H(1) \otimes H(1)}.$$

If $f: (H, d) \rightarrow (Q, t)$ is a **morphism** of such, then we denote

$$\mathfrak{R}(f) = f|_{H(1)}: \mathfrak{R}(H, d) \rightarrow \mathfrak{R}(Q, t).$$

Lemma

Let (H, d) , (Q, t) , and (R, s) be strictly graded braided Hopf algebras, then

1. $\mathfrak{R}(\text{id}_{(H, d)}) = \text{id}_{\mathfrak{R}(H, d)}$,
2. *for any morphisms of strictly graded braided Hopf algebras $f: (H, d) \rightarrow (Q, t)$ and $g: (Q, t) \rightarrow (R, s)$ it holds that*

$$\mathfrak{R}(g \circ f) = \mathfrak{R}(g) \circ \mathfrak{R}(f).$$

Lemma

Let (V, c) be a braided vector space and (H, d) a strictly graded braided Hopf algebra. Then for any braided linear map $f: (V, c) \rightarrow (H(1), \tilde{d})$ there exists a unique morphism of strictly graded braided Hopf algebras

$$\tilde{f}: \mathcal{B}(V, c) \rightarrow (H, d)$$

extending f , this is such that $i_H f = \tilde{f} p_V i_V$. In other words, such that the following diagram commutes;

$$\begin{array}{ccc}
 (V, c) & \xrightarrow{f} & (H(1), \tilde{d}) \\
 i_V \downarrow & & \downarrow i_H \\
 (T(V), \bar{c}) & & \\
 p_V \downarrow & & \\
 \mathcal{B}(V, c) & \xrightarrow{\tilde{f}} & (H, d).
 \end{array}$$

Theorem

Let (V, c) be a braided vector space, (H, d) a strictly graded braided Hopf algebra, and define

$$\begin{aligned} \varphi_{(V,c),(H,d)} : \operatorname{Hom}_{\operatorname{BrVec}}((V, c), \mathfrak{R}(H, d)) &\rightarrow \operatorname{Hom}_{\mathfrak{C}}(\mathcal{B}(V, c), (H, d)) \\ &: f \mapsto \tilde{f}. \end{aligned}$$

Then

- 1. $\varphi_{(V,c),(H,d)}$ is a bijection of sets,*
- 2. $\varphi_{(V,c),-} : \operatorname{Hom}_{\operatorname{BrVec}}((V, c), \mathfrak{R}(\bullet)) \rightarrow \operatorname{Hom}_{\mathfrak{C}}(\mathcal{B}(V, c), \bullet)$ is a natural transformation,*
- 3. $\varphi_{-, (H,d)} : \operatorname{Hom}_{\operatorname{BrVec}}(\bullet, \mathfrak{R}(H, d)) \rightarrow \operatorname{Hom}_{\mathfrak{C}}(\mathcal{B}(\bullet), (H, d))$ is a natural transformation.*

In conclusion, $\mathcal{B} \dashv \mathfrak{R}$ form a pair of adjoint functors.

Bibliography

- [1] N. Andruskiewitsch.
An introduction to Nichols algebras.
In *Quantization, geometry and noncommutative structures in mathematics and physics*, Math. Phys. Stud., pages 135–195. Springer, Cham, 2017.
- [2] N. Andruskiewitsch and M. Graña.
Braided Hopf algebras over non Abelian finite groups.
Bol. Acad. Nac. Cienc., Córdoba, 63:45–78, 1999.
- [3] V. G. Drinfel'd.
Quantum groups.
J. Sov. Math., 41(2):898–915, 1988.
- [4] I. Heckenberger and H.-J. Schneider.
Hopf algebras and root systems, volume 247 of *Math. Surv. Monogr.*
Providence, RI: American Mathematical Society (AMS), 2020.
- [5] C. Kassel.
Quantum groups, volume 155 of *Grad. Texts Math.*
New York, NY: Springer-Verlag, 1995.
- [6] S. Montgomery.
Hopf algebras and their actions on rings. Expanded version of ten lectures given at the CBMS Conference on Hopf algebras and their actions on rings, which took place at DePaul University in Chicago, USA, August 10-14, 1992, volume 82 of *Reg. Conf. Ser. Math.*
Providence, RI: American Mathematical Society, 1993.
- [7] M. E. Sweedler.
Hopf algebras.
New York: W.A. Benjamin, Inc. 1969, 336 p. (1969)., 1969.